

Ian Chi

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EDUCATION

- **University of California, Davis (B.S.) 2020-2024**
Physics and Mathematics (high honors), Cumulative GPA: 3.859
- **University of Santa Cruz, Davis (M.S.) 2025-Present**
Computer Science Engineering

SKILLS SUMMARY

- **Programming:**
 - Experienced: Python, Mathematica
 - Other: C, C#, Java, Carnap
- **Languages:** English, Chinese (Mandarin)
- **Others:** $\text{\LaTeX}$, LT Spice, Lean

PUBLICATIONS

- **Papers & Preprints:**
 1. I. Chi, T. Tan, M. Fraas. “**Poincaré disk as a model of squeezed states of a harmonic oscillator**”.
Paper Link: <https://pubs.aip.org/aip/jmp/article-abstract/66/6/062101/3349274>
Honors Thesis Link: www.math.ucdavis.edu/~webfiles/undergrad_thesis/202403_Ian_Chi_Fraas.pdf
 2. I. Chi, R. M. Morin, R. D. Mecholsky, N. A. Mecholsky. “**Derivation of Free Flight Distance Distribution with Application to Drift Velocity Calculations**”.
Poster: [Here](#)

AWARDS

- **Citation for Outstanding Performance, UC Davis Mathematics Department**
- **Citation for Outstanding Performance, UC Davis Physics Department**
- **Video Game Short Competition (AKA Game Jam) – Honorable Mention**
I worked as a designer for a small group and entered the GMTK Game jam 2023, a competition with 6000+ entrants. Won Honorable Mention with the game *SPARED!*. Link to list of games I have helped create: <https://splake975.itch.io/>
- **Dean’s List, School of Letters and Sciences – Fall 2020 & Spring 2021**

SELECTED COURSEWORK (UC DAVIS, B.S.)

- **Math courses**
 - Real Analysis, Modern Algebra, Topology, Complex Analysis, Euclidean Geometry, Probability Functional Analysis (Graduate), Algebraic Topology (Graduate),
- **Physics courses:**

- Classical Mechanics, Electricity and Magnetism, Quantum Mechanics, Electronics Lab, Particle Physics, General Relativity, Thermal and Statistical Mechanics, Computational and Mathematical,
- **Philosophy courses:**
 - Symbolic Logic, Philosophy of Mathematics, Philosophy of Science, Philosophy of Sexuality

SELECTED COURSEWORK (UC SANTA CRUZ, M.S.)

- **CSE**
 - Approximation Algorithms, Complexity Theory, Analysis Algorithms

EXTRACURRICULAR ACTIVITIES & EXPERIENCES

- **Directed Reading Program – Differential Topology and Geometry, (Fall 2022 to Winter 2023)**

A program offered by the Math department of UC Davis where I read through sections of the textbook *Introduction to Manifolds* by Tu, L. Read through sections on differential forms and co/tangent vectors/space, and manifolds. A powerpoint presentation was given at the end of the session. Done under the mentorship of Michael Ragone, PhD Candidate at UC Davis.

- **Directed Reading Program – Lie Groups and Lie algebras, (Spring 2023)**

Same program as above, performed under mentorship of Michael Ragone. Read through sections of *Introduction to Manifolds* by Tu, L. and *Lie Groups, Lie Algebras and Representations* by Hall, B.

- **2023 Physics Research Experience for Undergraduates (REU)**

Worked at Catholic University of America with Dr. Nicholas A. Mecholsky and Rachel M. Morin. see paper 2.

- **Labratory work developing new liquid scintillator mixtures for ANNIE and DUNE, Neutrino Experiments**

Working with Dr. Robert Svoboda, I have been in the lab working on new liquid scintillator materials that are projected to be used in the DUNE experiment. These materials are promised to be able to distinguish the light signals caused by Cherenkov radiation and scintillation by delaying the emission of scintillation light. These new materials are called Water Based Slow Scintillators (WBSS). My work involved testing how much and what new materials work in delaying the release of scintillation light. I also worked on creating a double coincidence light yield setup to test the scintillators.

- **Participant in the Research on Rails (Research Trip)**

Lead by Dr. Andrew Waldron, worked directly with PhD candidate Petr Vlachopoulos. The paper I read on this trip was *Topological quantum field theory* by Atiyah M. F. The paper is an introduction to TQFT and discusses goes over useful categories in regards to QFT. The category of manifolds of same dimension with morphisms being cobordisms and the category of vector spaces with arrows being linear maps are shown to be related with an interesting functor. the specifics of such a functor is useful in studying QFTs on non-flat spacetimes and the paper goes over several introductory examples of such interesting cases.

- **Self directed study group – QFT**

I studied quantum field theory with several other undergraduates and with Ben Guthrie, a prospective PhD student for Physics. The material we used was David Tong's, from the Perimeter Institute, online course

- **William Lowell Putnam Mathematical Competition – (December 2023)**

- **Nuclear Science and Security Consortium (NSSC) fellow** – (Spring 2023 to Summer 2024)
- **Advantaged Play (Casino) Software** – (February 2024 to Dec 2025)
Developed software used for beating casinos. Revenue in the 300k+ range and 7%+ returns over the period. Ended because of new regulations in the state of CA.